Part B

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Student No : 17.01.04.043

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Part B (A)

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Ans to the Ques No 1

There 13 instances, nine yes
four No

Entropy Measure

$$E(D) = \sum_{event \in class} -P(event)\log_2 P(event)$$

$$= -\{P(yes) \times \log_2 P(yes) + P(no) \times \log_2 P(no)\}$$

$$= -\{\tfrac{9}{13} \times \log_2 \tfrac{9}{13} + \tfrac{4}{13}\log_2 \tfrac{4}{13}\}$$

$$= 0.890$$

Outlook

$$E(outlook = sunny) = -(\tfrac{4}{5}\log_2 \tfrac{4}{5} + \tfrac{1}{5}\log_2 \tfrac{1}{5}) = 0.721$$

$$E(outlook = overcast) = -(\tfrac{3}{4}\log_2 \tfrac{3}{4} + \tfrac{1}{4}\log_2 \tfrac{1}{4}) = 0.811$$

$$E(outlook = raining) = -(\tfrac{2}{4}\log_2 \tfrac{2}{4} + \tfrac{2}{4}\log_2 \tfrac{2}{4}) = 1$$

$$E(outlook) = -(\tfrac{5}{13} \times 0.721 + \tfrac{4}{13} \times 0.811 + \tfrac{4}{13} \times 1)$$

$$= 0.834$$

$$Gain = E(D) - E(outlook) = 0.056$$

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<u>Temp</u>

$E(\text{temp}=\text{hot}) = -\left(\frac{2}{4}\log_2\frac{2}{4}+\frac{2}{4}\log_2\frac{2}{4}\right) = 1$

$E(\text{temp}=\text{mild}) = -\left(\frac{4}{5}\log_2\frac{4}{5}+\frac{1}{5}\log_2\frac{1}{5}\right) = 0.721$

$E(\text{temp}=\text{cool}) = -\left(\frac{3}{4}\log_2\frac{3}{4}\right)+\frac{1}{4}\log_2\frac{1}{4}\right) = 0.811$

$E(\text{temp}) = -\left(\frac{4}{13}\times 1+\frac{5}{13}\times 0.721+\frac{4}{13}\times 0.811\right)$

$= 0.834$

$\text{Gain} = E(D) - E(\text{temp}) = 0.056$

<u>Windy</u>

$E(\text{Windy}=\text{True}) = -\left(\frac{3}{5}\log_2\frac{3}{5}+\frac{2}{5}\log_2\frac{2}{5}\right)$

$= 0.971$

$E(\text{windy}=\text{false}) = -\left(\frac{6}{8}\log_2\frac{6}{8}+\frac{2}{8}\log_2\frac{2}{8}\right)$

$= 0.811$

$E(\text{windy}) = -\left(\frac{5}{13}\times 0.971+\frac{8}{13}\times 811\right)$

$E(\text{windy}) = 0.872$

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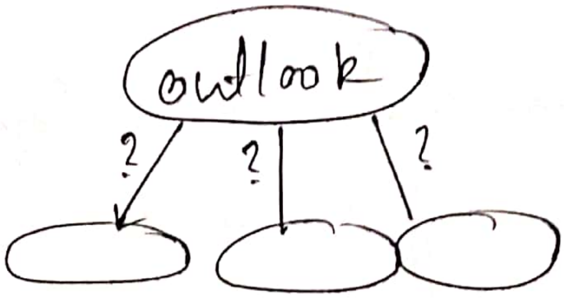
Gain = E (D) − E (windy) = 0.018

So, Gain (outlook is highest) = 0.056



So, root is outlook.

outlook	temp	windy	Play
sunny	hot	True	no
Sunny	hot	False	Yes
sunny	Cool	True	Yes
sunny	mild	false	Yes
sunny	hot	True	Yes.

$$E(D) = -\left(\frac{4}{5}\log_2\frac{4}{5} + \frac{1}{5}\log_2\frac{1}{5}\right) = 0.721$$

$\boxed{\text{temp}}$

$$(\text{temp} = \text{hot}) = -\left(\frac{2}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{1}{3}\right)$$

$$= 0.918$$

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$(temp = cool) = -(\frac{1}{1}\log_2\frac{1}{1} + 0) = 0$

$(temp = mild) = -(\frac{1}{1}\log_2\frac{1}{1} + 0) = 0$

$E(temp) = -(\frac{3}{5} \times 0.918 + \frac{1}{5} \times 0 + \frac{1}{5} \times 0)$

$= 0.550$

$Gain = E(D) - E(temp) = 0.721 - 0.550$

$= 0.71$

$\boxed{Winty}$

$(winty = True) = -(\frac{2}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{1}{3})$

$= 0.918$

$(winty = false) = -(\frac{2}{2}\log_2\frac{2}{2} + 0) = 0$

$E(windy) = -(\frac{3}{5} \times 0.918 + 0) = 0.550$

$\therefore Gain = E(D) - E(temp) = 0.721 - 0.550$

$= 0.171$

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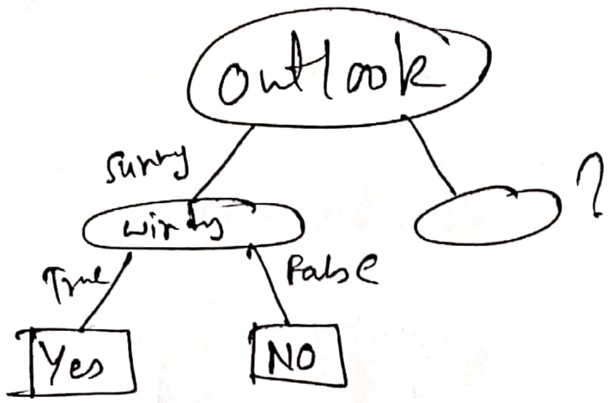
So, both are same again, Last take windy so the tree is now.



Outlook	Temp	Play
overcast	hot	no
overcast	cool	Yes
overcast	cool	yes
overcast	mild	yes

$$E(D) = -\left(\frac{3}{4}\log_2\frac{3}{4} + \frac{1}{4}\log_2\frac{1}{4}\right) = 0.811$$

$$(temp = hot) = 0$$

$$(temp = cool) = -\left(\frac{2}{2}\log_2\frac{2}{2} + 0\right) = 0$$

$$(temp = mild) = -\left(\frac{1}{1}\log_2\frac{1}{1} + 0\right) = 0$$

$$\therefore E(temp) = 0 \times 0.811 = 0$$

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As there is no other ~~#~~ gain and all values are zero lets take cool as subclass.

Now lets see about rain.

Outlook	temp	play
rainy	mild	yes
rainy	mild	yes
rainy	cool	no
rainy	mild	no

$$E(D) = -\left(\frac{2}{4}\log_2\frac{2}{4} + \frac{2}{4}\log_2\frac{2}{4}\right) = 1$$

$$(\text{rainy}=\text{mild}) = -\left(\frac{2}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{1}{3}\right) = 0.918$$

$$(\text{rainy}=\text{cool}) = 0$$

$$E(\text{rainy}) = -\left(\frac{3}{4}\times 0.918\right) + \frac{1}{4}\times 0) = 0.688$$

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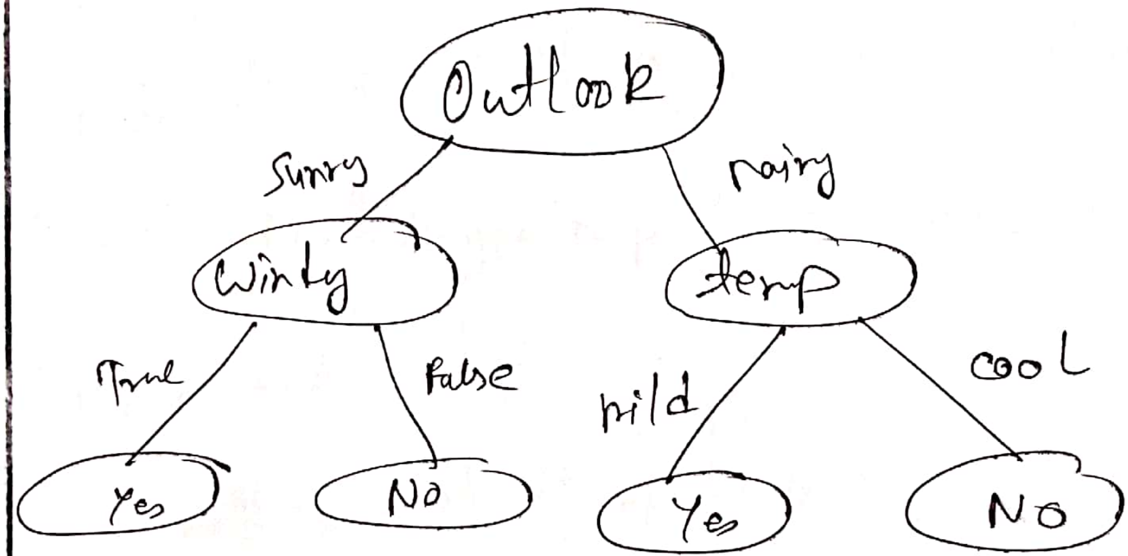
So, clearly hain of rain is greater to its selected So the tree is Now.



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Ans to the Qus No 2 (a)

for a multivariate dataset,

$$\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + \dots + b_n x_n$$

This is the regression line based on the sample data.

The method of least squares choose the value of $b_1, b_2, \dots b_n$ to minimize the sum of squared errors.

$$SSE = \sum_{i=1}^{m} (y_i - \hat{y}_i)^2$$

$$= \sum_{i=1}^{m} (y - b_0 - b_1 x_1 - b_2 x_2 - \dots - b_n x_n)$$

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Using calculus,
We obtain the following formula:

$$b_1 = \frac{\sum_{i=1}^{m} (x_1^i - \bar{x}_1)(y_i - \bar{y})}{\sum_{i=1}^{m} (x_1^i - \bar{x}_1)}$$

$$= \frac{m \sum_{i=1}^{m} x_1^i y_i - \sum_{i=1}^{m} x_1^i \sum_{i=1}^{m} y_i}{m \sum_{i=1}^{m} x_1^{i2} - \left(\sum_{i=1}^{m} x_1^i\right)^2}$$

Or $b_1 = r \frac{S_y}{S_{x_1}}$

$\vdots$

$$b_n = \frac{\sum_{i=1}^{m} (x_n^i - \bar{x}_n)(y_i - \bar{y})}{\sum_{i=1}^{m} (x_n^i - \bar{x}_n)}$$

$$= \frac{m \sum_{i=1}^{m} x_n^i y_i - \sum_{i=1}^{m} x_n^i \sum_{i=1}^{m} y_i}{m \sum_{i=1}^{m} x_n^{i2} - \left(\sum_{i=1}^{m} x_n^i\right)^2}$$

Or, $b_n = r \frac{S_y}{S_{x_n}}$

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Again, $b_0 = \overline{y} - b_1 \overline{x_1} - b_2 \overline{x_2} - \dots - b_n \overline{x_n}$

Soln

A = 12

$\therefore$ Data Points: (12, 13), (13, 14), (15, 18), (16, 16), (18, 22)

Here,

x	y	xy	x^2
12	13	156	144
13	14	182	169
15	18	270	225
16	16	256	256
18	22	396	324
$\Sigma x = 74$	$\Sigma y = 83$	$\Sigma xy = 1260$	$\Sigma x^2 = 1118$

$$\therefore b_1 = \frac{(5 \times 1260) - (74 \times 83)}{(5 \times 1118) - (74)^2} = \frac{158}{114} = 1.386$$

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Now, $\bar{Y} = \frac{83}{5}$, $\bar{x} = \frac{74}{5}$

$= 16.6$ $\qquad = 14.8$

$\therefore b_0 = 16.6 - 1.386 \times 14.8$

$= -3.9128$

$\therefore$ Regression line:

$$y = -3.9128 + 1.386x$$

This is the the best fitting line eqn.

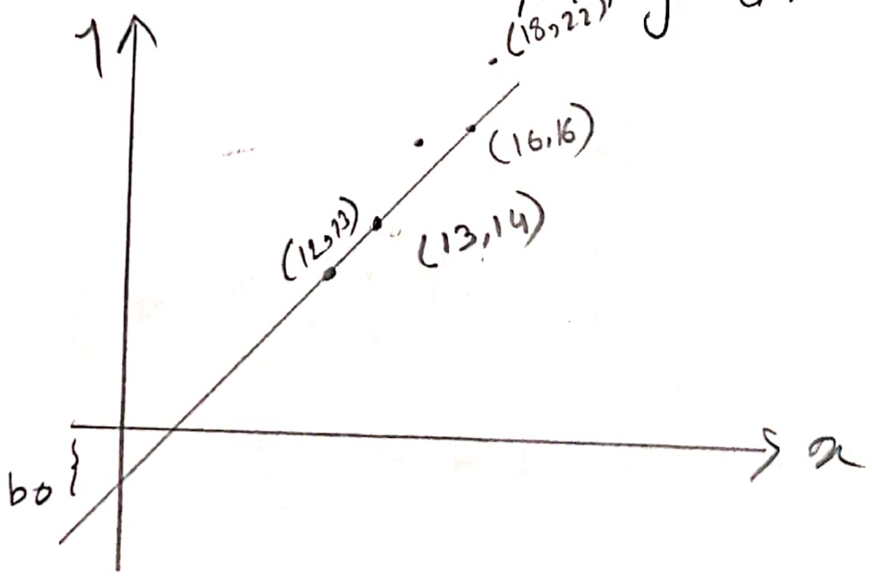

fig : Best-Fitted Regression line.

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Ans to the Ques NO 2 (b)

Example

Actual(Y)	Predicted ($\hat{Y}$)	Error ($E1 = Y - \hat{Y}$)	$E1^2$	$E2 = Y - \bar{Y}$	$E2^2$
10	10	0	0	-10	100
20	20	0	0	0	0
30	30	0	0	10	100
Mean = 20		Sum = 0	Sum = 0		Sum = 200

$$R\text{-squared} = 1 - \frac{S_{res}^2}{S_{total}^2}$$

$$= 1 - \frac{0}{200} = 1$$

$$MAE = \frac{1}{N} \sum_{i=1}^{N} |\hat{Y} - Y_i| \quad , \quad \hat{Y} = \text{Prediction}, \; Y_i = \text{True value}$$

$$= \frac{1}{3} \times E1 = \frac{1}{3} \times 0 = 0$$

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$$MSE = \frac{1}{N}\sum_{i=1}^{N}(\hat{Y} - Y_i)^2$$

$$= \frac{1}{3} \times E_1^2 = \frac{1}{3} \times 0 = 0$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{Y} - Y_i)^2}$$

$$= \sqrt{\frac{1}{3} \times E_1^2} = \sqrt{0} = 0$$

The categorical Naive Bayes classifier is suitable for classification with discrete features that are categorically distributed.

The categories of each feature are drawn from a categorical distribution.

Thus we can use categorical Naive Bayes classifier for categorical valued dataset.

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Ans to the Ques No 1

(b) Given

a binary dataset contains five different continuous features based on real life scenario.

Assumption:

(i) All the data are evenly distributed between two distinct classes.

which means,

half of the data belong to another class.

(ii) All the data are of one class,

Let $n=10$

$$\therefore \text{Entropy}(D) = -\left(\frac{10}{10}\log_2\frac{10}{10} + \frac{0}{10}\log_2\frac{0}{10}\right) = 0$$

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Here, we can see that entropy is zero, which means the absence of entropy.

Hence,

gain(D, A1) = Entropy(D) − E(A1) = −E(A1)

gain(D, A2) = Entropy(D) − E(A2) = −E(A2)

gain(D, A3) = Entropy(D) − E(A3) = −E(A3)

gain(D, A4) = Entropy(D) − E(A4) = −E(A4)

gain(D, A5) = Entropy(D) − E(A5) = −E(A5)

Let n = 10

$\therefore$ Entropy(D) = $-\left(\frac{5}{10}\log_2\frac{5}{10} + \frac{5}{10}\log_2\frac{5}{10}\right)$

= 1

Here we can see that entropy is maximum which is +1

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Here,

For five att,

$gain(D, A_1) = Entropy(D) - E(A_1)$

$gain(D, A_2) = Entropy(D) - E(A_2)$

$gain(D, A_3) = Entropy(D) - E(A_3)$

$gain(D, A_4) = Entropy(D) - E(A_4)$

$gain(D, A_5) = Entropy(D) - E(A_5)$